E-Logbook requirements

Purpose:

The Electronic Logbook module aims to digitize plant operations logging, replacing paper-based shift logs with a structured, real-time, configurable, and auditable platform. This ensures smooth shift handovers and operational traceability in pulp and paper manufacturing facilities. The module empowers plant managers/admin to create, configure and customize logbook for different usecases accordingly.

Target Audience:

- Operators
- Shift Supervisors
- Lab engineers
- Maintenance Team
- Plant Managers

Figma design:

(Redesign) Mt. Fuji

Target audience:

A. Logbook Admin (Process Engineer/Admin User)

> **Primary Goal:** Configure, customize, and manage logbooks for different processes.

B. Operator (Plant Operator/Shift Operator)

> **Primary Goal:** Log accurate process data and events in a timely and efficient manner during shifts.

C. Shift Incharge/Shift Engineer

> **Primary Goal:** Oversee shift operations, ensure task completion, and facilitate accurate shift handovers.

User stories:

Story 1) As an admin, I want to view a list of all logbooks and create multiple logbooks as per operational needs.

Description:

The Logbook module landing page enables admins to:

- View a list of all created logbooks.
- Clicking on a particular logbook row will open the 'View Log History' page, which displays all log entries shift-wise. Admins can view previous logs from this page.
- Search for logbooks by name.
- Filter logbooks by plant or facility.
- Create new logbooks using a modal form.
- Interact with logbooks through an overflow menu with options like:
  - View log history
  - Edit and configure
  - Pin
  - Delete

Acceptance Criteria:

SNo	Criteria	Description
1	View logbooks	Admin sees a list of all logbooks in a table with columns: Name, Description, Plant, Facility, Created By, Last Updated.
2	Search by name	Typing in the search bar filters the list by logbook name in real time.
3	Filter logbooks	Dropdown filters for Plant and Facility.
4	Create logbook	Clicking "Create new logbook" opens a modal where the admin can input Name, Description, Plant, and Facility.

5	Action menu	Each logbook row has a 3-dot menu with: View log history, Edit & configure, Pin, and Delete.
6	Validation	All fields in the logbook creation modal are mandatory. Errors are shown for empty/invalid inputs.
7	Pinned item display	Pinned logbooks are visually highlighted (pinned icon) and shown before the logbook name.
8	Empty state	If no logbooks exist or no results match the search/filter, show a clean empty state message.
9	Pagination	Logbooks are paginated (default 10/page); navigation at the bottom.

Search & Filter Details:

Search:

- Text field (top of the table).
- Case-insensitive, match search on logbook Name.

Filter:

- Filter by Plant (dropdown, multi-plant support)
- Filter by Facility (dropdown, dynamic based on selected plant if needed)
- Filters are optional and can be cleared.

Empty State Handling

Case	UI Message
No logbooks created	“No logbooks found. Click ‘Create new logbook’ to get started.”
No results on search/filter	“No logbooks match your search or filter criteria.”

Layout and Usability

- **Primary CTA** (“Create new logbook”) is top-right, visually prominent.
- **Search bar**
- **Filters** are dropdowns aligned to the right of the search bar.
- **Table design** is clean with sortable headers (Name, Plant, Last updated).
- **Pinned logbooks** have a distinct icon and positioning (top of list).

- **Pagination** with total count indicator (e.g., "Showing 2 of 2 logbooks").

Validation Checks in “Create New Logbook” Modal:

Field	Validation Rule	Error Message
Name	Required, max 40 chars, no special chars	"Please enter a valid logbook name."
Description	Required, max 100 chars	"Please provide a short description."
Plant	Required (dropdown must have selection)	"Select a plant."
Facility	Required (dropdown must have selection)	"Select a facility."

- Submit button is disabled until all validations pass.
- Duplicate logbook names should trigger: “Logbook name already exists.”

Story 2) [As an admin, I can configure logbook according to my use cases.]

Description

Once the Admin completes the initial step of creating a logbook by filling out the required four fields (Logbook Name, Description, Plant, and Facility), they are redirected to the configuration screen. Here, they can build and customize the logbook form by adding fields such as:

- **Text**
- **Checklist**
- **Dropdown**
- **Paragraph (Long text)**
- **Attachments**
- **Table**

Each field added to the canvas has:

- A Configure CTA to set its label and values.
- A Delete CTA to remove the element.

For the Table element, admins must choose a Logging Mode:

- Fixed Interval

- Sampling Based

Depending on the selected logging mode, only relevant parameters will be shown:

- **Fixed Interval mode**: Requires setting start time, end time, and interval. Only Fixed Interval parameters and Sensor Tag parameters are shown when the user clicks "Add Parameters."
- **Sampling Based Mode:** No time inputs are required. Only Sampling Based parameters and Sensor Tag parameters are shown when the user clicks "Add Parameters."

This segregation prevents parameter-type mismatch and ensures structured data logging.

Once all fields are configured:

- **Preview**
  - Admin clicks the **Preview** button to view the logbook layout as it would appear to users.
  - All configured fields display with placeholder values.
  - Admin can test interactivity like dropdowns, checklists, etc.
  - No data is saved in preview mode.
- **Publish**
  - Once satisfied, admin clicks **Publish**.
  - On success, the logbook is saved and made available in the Logbook module.

Acceptance Criteria

1. **Redirect on Logbook Creation**
   - Admin is redirected to the configuration screen immediately after successful creation.
2. **Field Addition & Customization**
   - Admin can add multiple fields from the right panel (text, checklist, dropdown, long text, attachments, table).
   - Each field has:
     - A "Configure" button that opens a side panel with appropriate settings.
     - A "Delete" icon to remove the element from canvas.
3. **Field Configuration**
   - **Text, Long Text**: Require only a label.
   - **Checklist, Dropdown**: Require label and at least one option (see screenshots).
   - **Table**:
     - Admin must select logging mode: Fixed Interval or Sampling Based.
     - Based on logging mode, only valid parameter types are shown.

    - **Fixed Interval** additionally requires:
        - Start time, end time, and interval.

4. **Table Parameter Filtering**
    - On **Fixed Interval**, only show Fixed Interval Parameters and Sensor Tags.
    - On **Sampling Based**, only show Sampling Based Parameters and Sensor Tags.

5. **Preview mode:** Opens full-screen or modal preview of logbook layout.

6. **Publish flow:** Succeeds and redirects back to Logbook overview/list.

Empty State Handling

- **Initial Configuration Page (no fields added):**
    - Get started by adding fields from the right panel
- **Dropdown/Checklist Fields with No Items:**
    - Show warning: “At least one option must be added.”
- **Table with No Parameters:**
    - Show inline warning: “Please add at least one valid parameter.”

Layout and Usability

- **Right-Side Panel** for adding and configuring fields ensures separation of action and content.
- **Sticky top bar** with `Preview` and `Publish` buttons ensures easy access to key actions.
- Each field on the canvas clearly displays:
    - Label (once configured)
    - Configure icon (gear)
    - Delete icon (trash)
- For table logging:
    - Toggle for selecting logging mode.
    - Auto-refresh parameter list based on mode.
- Smooth transitions between field configuration and canvas updates.

Validation Checks

- **General**
    - Every added field must have a label.
    - Dropdown and Checklist must have at least one item.
    - Cannot publish the logbook without at least one configured field.

- Prevent publishing if required configurations (like table logging mode or missing time range) are incomplete.
- **Table Field**
  - If Fixed Interval is selected:
    - Validate that start time < end time.
    - Interval must be > 0 and fit within the time range.
  - At least one valid parameter must be added.
  - Show only parameter types relevant to the selected logging mode.

Story 3) [As an admin, I can create manual tags and assign a logging mode to each manual tag.]

Description

User can create manual tags that will later be used in logbooks. During creation, the admin must assign a logging mode to each tag, which determines how it can be used in logbook configurations.

There are two supported logging modes:

- **Fixed Interval**
- **Sampling Based**

These modes directly affect:

- Where the tag can be used (e.g., which tables in logbooks).
- How the data for that tag is expected to be entered/logged.

Story 4) [As a user(Admin/Operator), I can view past logbooks by selecting the date and shift.]

Description

- This functionality allows both Admin and Operator users to access historical logbooks based on specific date, shift, and logged by filters.
- There will be an additional 'Comments' column with an icon that displays the operator's logged comments on hover. @Rahul Pagar requesting you to add it in the design too.
- It ensures visibility into previously recorded operational data for review, audits, and reference.
- Admins can access the log history from the Logbook landing page either by clicking a row entry or by using the three-dot menu beside each logbook. Operators, however, can only access the same page via the three-dot menu.

- All historical logs are viewable in a shift-wise layout with date and user who logged.
- Logs can be edited within 7 days; post that, they become non-editable.

Acceptance Criteria

SNo	Criteria
1	Admin can click on a logbook row to access past log history, and can also access the same via the 3-dot menu and selecting "View Log History."
2	Operator can access the log history via the 3-dot menu and selecting "View Log History."
3	Logbook history page displays all entries in a structured date-wise manner consisting of the shift details, Last updated time and the operator who logged, and comments.
4	User can filter the list of past logbooks by date, shift, and logged by.
5	User can search specific logbooks using comment text.
6	Entire row in the listing is clickable and redirects to the selected logbook.
7	Edit option is available only if the logbook entry is less than 7 days old. After 7 days, entries become read-only.
8	Admins and Operators both see the same UI on the detailed view page.

Search and Filter Details

- **Search**:
  - Input: `Search comments`
  - Use: Free-text search; matches keywords inside long text field section.
  - Behavior: Partial match results highlighted; clicking result opens detailed logbook view.
- **Filters**:
  - **Date Picker**: Choose the day for which logs are to be viewed.
  - **Shift Selector**: Dropdown with values `Shift 1`, `Shift 2`, `Shift 3`.
  - **Logged By**: Dropdown with names of operators or admins who filled the log.
- **Sorting**:
  - Columns `Date`, `Shift`, `Last updated time`, and `Logged by` are sortable in ascending/descending order.

Empty State Handling

Scenario	Message/Visual
No logbooks for selected filters	“No logbooks available for the selected filters.”
No results for search	“No comments matched your search. Try another keyword.”
No logs created yet	“No logs have been entered yet”
No comments logged by the operator.	“No comments logged”

Layout and Usability

Landing Page (Past Logs View):

- Table with columns: `Date`, `Shift`, `Last updated time`, `Logged by`, `comments icon`
- Entire row is clickable for both admin and operator
- Search and filter controls placed above the table for accessibility

Detailed View Page (Per Shift):

- Displays all configured fields: tables (fixed/sampling), checklists, text, attachments
- Structured format based on configuration
- Edit button on top-right (visible and enabled only for <7-day entries)
- For >7 days: show info tooltip stating: “This logbook is beyond the editable period.”

Validation Checks

Field	Validation
Filters	Ensure valid date selected; shift and logged-by are optional
Comment search	Minimum 1 word for search to initiate
Edit button visibility	Based on log entry date; disable if older than 7 days
Redirect	Ensure clicking row or search result correctly opens the detailed log

Story 5) [As an admin, I want to automatically generate shift-wise logbook reports to relevant stakeholders, such as plant managers, so they stay informed.]

Description

- This feature allows admin users to configure automated email reports for logbooks.
- The reports can be scheduled shift-wise or daily, and recipients can be selected from the list of Mt. Fuji users within that plant.
- This helps stakeholders stay informed about plant operations without manual follow-ups.
- Each logbook will have its own individual report settings.

Acceptance Criteria

SNo	Criteria
1	Admin can access the "Generate Report" popup from the logbook 3 dot menu and selecting “Report generation settings”.
2	Admin can select the report frequency from a dropdown — options include `Shift-wise` and `Daily`.
3	Admin can select one or more recipients from a dropdown list of Mt. Fuji users of that plant.
4	Report settings are saved on clicking “Save” and cancelled on clicking “Cancel.”
5	If either field is left unfilled, an error should appear prompting the user to complete the missing fields.
6	Once configured, reports are automatically sent via email to the recipients with shift or daily summaries depending on the selected frequency.

Empty State Handling

- **Before configuration**: No reports are sent; admin must explicitly configure reports.
- **Empty Recipients**: If no recipients are selected, display: “Please select at least one recipient.”
- **Empty Frequency**: If no frequency is selected, display: “Please select a report frequency.”

Layout & Usability

- Admin can access the "Generate Report" popup from the logbook 3 dot menu and selecting “Report generation settings”.

- Clicking the “Report generation settings” opens a **modal/popup** with:
  - Dropdown: `Report Frequency` (Shift-wise, Daily)
  - Dropdown: `Recipients` (multi-select from user list of that plant)
  - Action buttons: `Cancel`, `Save`
- Modal closes after saving and shows a confirmation snackbar: “Report configuration saved successfully.”

Validation Checks

Field	Validation
Frequency	Must be one of: `Shift-wise`, `Daily`
Recipients	At least one recipient must be selected
Save Button	Disabled until both fields are filled
Cancel Button	Always active, closes modal without saving

Email Content

1. Report frequency- Shiftwise

- **Subject**: Shift Log Report – [Logbook Name] | [Shift Name], [Date]
- **Body**:
  Hi [Name],

  Here’s the shift log report for **[Logbook Name]** – **[Shift Name]**, dated **[Date]**.
  - **Logged by:** [Operator Name]
  - **Logbook:** [Logbook Name]
  - **Shift:** [Shift Name]

  Click below to view the complete report:
  [Attachment]

  Best regards,
  Mt. Fuji Team

2. Report frequency- Daily

   Subject: Daily Log Report – [Logbook Name] | [Date]

   Body:

Hi [Name],

Here's the **daily shift log report** for **[Logbook Name]**, compiled for **[Date]** across all shifts.

Logbook: [Logbook Name]
Date: [Date]
Shift-wise Summary:

- **Shift 1** – Logged by: [Operator Name 1]
- **Shift 2** – Logged by: [Operator Name 2]
- **Shift 3** – Logged by: [Operator Name 3]

Click below to view the complete reports:
[Attachments]

Best regards,
Mt. Fuji Team

Story 6) [As an operator, I want to view the landing page of the logbook, select a particular logbook and shift, and start logging entries.]

Description:

- The landing page of Operator shows a list of logbooks accessible to them.
- Operator should see only those logbooks for which they have facility-level editor access. For example, if the operator has editor access to the "Paper Machine" facility, only logbooks from this facility will be visible.
- They can search the logbook by name or filter the logbook list by Plant and Facility.
- Interact with logbooks through the 3 dot menu with options like:
- View log history
- Pin
- On selecting/clicking a logbook row:
  - A modal prompts the user to select a shift (only current or past).
  - "Start Logging" button is enabled only after shift selection.
- Once a shift is selected and logging is initiated:
  - Operator's name and selected shift details should be autofetched and shown at the top.
  - The editable logbook form for that shift opens.
- Auto-fetch CTA is visible only next to sensor tags, allowing users to autofill values captured from sensors at specified times.

- The operator can manually input the entry fields visible on the form.
- Inline validation is enforced for:
  - Data type (e.g., numbers)
  - Min-Max bounds
  - Required fields
- Final submission is done via "Save & Update" CTA. Clicking it saves the data to the database and updates the logbook status.
- Empty state message is shown if no logbooks are available for their access scope.

Acceptance Criteria:

SNo	Criteria Description
1	Operator should only see logbooks for which they have editor access at the facility level.
2	If no logbooks are available to the operator, show the message: "No logbooks available for your role. Please contact your supervisor or system admin for access."
3	Operator can search logbooks by logbook name.
4	Operator can filter logbooks by Plant and Facility.
5	Operator can access additional options from 3-dot menu: View Log History, Pin.
6	On clicking a logbook, a modal appears to select a shift. Only current or past shifts can be selected (future shifts disabled).
7	The "Start Logging" button is enabled only after a valid shift is selected.
8	Once shift is selected, Operator’s name and selected shift appear at the top of the logging screen.
9	The editable form for that shift is displayed.
10	An Auto-fetch CTA appears only next to sensor tags. On click, it fills values captured at the defined time.
11	Auto-fetched sensor tag values cannot be edited.
12	Operator can manually input values for non-sensor fields.

13	All required fields must be filled before saving.
14	Fields must match the expected data type (e.g., numeric, text).
15	Tags with min/max bounds should enforce limits.
16	Logged values should align with shift timings and defined intervals.
17	“Save & Update” button is disabled until all validations are passed.
18	On clicking “Save & Update”, the data is saved to the database and logbook status is updated.

Empty State Handling:

- If no logbooks are available to the operator (based on access role or filters), display:

  “No logbooks available for your role. Please contact your supervisor or system admin for access.”

Layout and Usability:

- Home page: List of available logbooks with search/filter bar on top.
- Shift selection modal (on click).
- Shift selection modal:
  - Dropdown to select from valid shifts (disabled for future dates).
- Logging screen:
  - Auto-fetch buttons clearly aligned with sensor tags only.
- “Save &Update “ CTA: Disabled by default until all validation checks are passed.

Validation Checks:

- Required Fields: All mandatory fields must be filled in.
- Data Type: Accept only the expected formats for tag values, as defined during their creation.
- Value Range: Enforce minimum and maximum constraints for tags, if specified.
- Sensor Tags: Prevent editing of sensor tag values if they are auto-fetched.
- Time-specific Data: Ensure values align with shift timings and fixed intervals.
- Form Submission: Enable submission only when all the above conditions are satisfied.

Data formats:

Data type	Format(UI Control)	Example (Usecases)
Number	Numeric input field	Temp = 42.5 C, Speed of motor= 15.22 m/s
Text(free form)	Text area (Multi-line)	Pump number 3 vibrating loudly.
Date	Calendar	2025-04-24
Date & Time	Calendar + Time picker	2025-04-24 “14:02:56”
Checkbox	Toggle (Yes/No)	☑ Safety valve inspected ☐ Pump inspection
Dropdown	predefined list	Status = Running, Stopped, Fault
Media attachment	Image upload	Pump leak image

Accessibility:

Plant level access(Configurator access)

Actions	Admin	Editor	Viewer
Create new logbook	☑	☐	☐
Edit and configure template	☑	☑	☐
Pin/Unpin	☑	☑	☑
Delete logbook	☑	☐	☐
Edit log history	☑	☑	☐
View log history	☑	☑	☑

Facility level access(Operator access)

Actions	Admin	Editor	Viewer
Create new logbook	☐	☐	☐
Edit and configure template	☐	☐	☐
Pin/Unpin	☑	☑	☑
Delete logbook	☐	☐	☐
Edit log history	☑	☑	☐
View log history	☑	☑	☑

Success Metrics:

- Successful delivery of the Logbook module to both of our currently onboarded clients, i.e., ANC and Action Tesa.
- Positive client feedback post-implementation.
- Consistent usage of the Logbook feature over a 4-week period, with shift-wise entries logged as per plant norms, will lead to repeat engagement and an increase in the number of users onboarded onto Mt. Fuji.

Future considerations:

- More number of fields will be added to get data from operator in more easy ways and to make it interactive,
- Gen AI integration
- Activity storing for traceability

Tech team discussions:

- For auto-fetching sensor tag data, we will provide a CTA (Call to Action) to auto-fetch values. On click, the values for that tag will be auto-populated and will not be editable.
- Move the "Generate Report" entry point to the 3-dot menu instead of keeping it within the configuration section.

- Create an export service to download shift-wise logbook reports as PDFs and send them via email to decision-makers.
- Integrate a search comments feature that scans the text provided by the user in long text fields (comments) and returns a list of relevant logbooks where the text exists.